METRIC

MIL-DTL-24581A(NAVY)
18 December 2018

SUPERSEDING
DOD-A-24581(NAVY)
25 October 1978

DETAIL SPECIFICATION

ANCHOR, NAVY STOCKLESS TYPE

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers nonfouling, Navy stockless type anchors for use on ships.

1.2 Classification. The anchors are of the following types, classes, and nominal masses, as specified (see 6.2).

1.2.1 Types and classes.

a. Type I – Anchors having a mass between 50 kilograms (kg) (100 pounds [lbs]) and 22,500 kg (45,000 lbs) inclusive.

(1) Class 1 – Without a stabilizer

(2) Class 2 – With a stabilizer

b. Type II – Anchors having a mass of 30,000 kg (60,000 lbs) without a stabilizer.

1.2.2 Nominal masses.

Nominal masses kg (lbs)		
50 (100)	800 (1,600)	5,000 (10,000)
100 (200)	850 (1,700)	5,500 (11,000)
150 (300)	900 (1,800)	6,000 (12,000)
200 (400)	950 (1,900)	6,500 (13,000)
250 (500)	1,000 (2,000)	7,250 (14,500)
300 (600)	1,100 (2,200)	7,500 (15,000)
350 (700)	1,250 (2,500)	8,000 (16,000)
400 (800)	1,500 (3,000)	9,000 (18,000)
450 (900)	1,750 (3,500)	10,000 (20,000)
500 (1,000)	2,000 (4,000)	11,250 (22,500)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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Nominal masses - continued		
550 (1,100)	2,500 (5,000)	12,500 (25,000)
600 (1,200)	3,000 (6,000)	15,000 (30,000)
650 (1,300)	3,500 (7,000)	17,500 (35,000)
700 (1,400)	4,000 (8,000)	20,000 (40,000)
750 (1,500)	4,500 (9,000)	22,500 (45,000)
		30,000 (60,000)

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-22200/1 - Electrodes, Welding, Mineral Covered, Iron-Powder, Low-Hydrogen Medium and High Tensile Steel, As-Welded or Stress-Relieved Weld Application

MIL-PRF-23236 - Coating Systems for Ship Structures

MIL-PRF-24635 - Coating Systems, Weather-Resistant, Exterior Use

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-2035 - Nondestructive Testing Acceptance Criteria

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL FACILITIES (NAVFAC) ENGINEERING COMMAND DRAWINGS

620656 - Standard Fleet Moorings, Stabilizer Details for Navy Standard Stockless Anchors

3141650 - Anchor, Navy Stockless Type, Stabilized

(Copies of these documents are available from NAVFAC EXWC, 1000 23rd Avenue, Port Hueneme, CA 93043.)

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NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-860337 - Stockless Anchors Assembly and Details (50 kg to 22,500 kg)

803-920808 - 30,000 kg Stockless Anchor, Mark II, Assembly and Details

(Copies of these documents are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of these documents may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9074-AQ-GIB-010/248 - Requirements for Welding and Brazing Procedure and Performance Qualification

T9074-AS-GIB-010/271 - Requirements for Nondestructive Testing Methods

(Copies of these documents are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ [by searching for the document number without the suffix]. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A27/A27M - Standard Specification for Steel Castings, Carbon, for General Application

ASTM A36/A36M - Standard Specification for Carbon Structural Steel

ASTM A53/A53M - Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

ASTM A370 - Standard Test Methods and Definitions for Mechanical Testing of Steel Products

ASTM A668/A668M - Standard Specification for Steel Forgings, Carbon and Alloy, for General Industrial Use

ASTM E112 - Standard Test Methods for Determining Average Grain Size

(Copies of these documents are available online at www.astm.org.)

NACE INTERNATIONAL

NACE No. 2/SSPC-SP 10 - Near-White Metal Blast Cleaning

NACE WJ-2/SSPC-SP WJ-2 - Waterjet Cleaning of Metals – Very Thorough Cleaning (WJ-2)

(Copies of these documents are available online at www.nace.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Material.

3.1.1 Anchor crown and flukes. Anchor crown and flukes shall be steel castings in accordance with ASTM A27/A27M. Castings shall be heat-treated by normalizing and tempering to obtain the mechanical properties listed in table I. The detailed procedures to be used shall be determined by the manufacturer. The microstructure specimen shall have a minimum grain size of 6 in accordance with ASTM E112. Material having a grain size number less than 6 (larger grain size) shall be subject to rejection.

TABLE I. Mechanical properties.

Mechanical properties	Anchor mass	
	50 to 15,000 kg	Over 15,000 kg
Tensile strength	4,218 kg/cm^2 (60,000 lb/in^2)	4,921 kg/cm^2 (70,000 lb/in^2)
Yield point	2,109 kg/cm^2 (30,000 lb/in^2)	2,812 kg/cm^2 (40,000 lb/in^2)

3.1.1.1 Heat treatment. In addition to the requirements of ASTM A27/A27M, castings may be reheat-treated by austenitizing, quenching, and tempering, or normalizing and tempering as necessary to attain the required physical properties.

3.1.1.2 Crown and fluke test block. A test block shall be integrally cast with the crown and flukes (unless damage to parts could result, in which case a separate cast test block from the heat poured with the casting shall be required) to comply with the chemical, mechanical, and microstructure test requirements. Adequate measures shall be taken to ensure test block soundness. Chills or chill molds shall not be used for the test block. The size of the test block shall be in proportion to the thickness (T), where T is the diameter of the largest sphere (±12.7 millimeters [mm] [±0.50 inch]) that can be inscribed in any cross section of the casting. Test blocks shall remain attached to the casting they represent until submitted for inspection. Test blocks shall be heat-treated with the casting. The minimum dimensions of the test block shall be T by T by 2T. Minimum T shall be 101.6 mm (4 inches) for all size anchors. Dimensions in excess of these shall be permitted providing sound test blocks are produced. Mechanical testing (see 4.5.24.4.2) shall be taken at least 25.4 mm (1 inch) below the test block surface.

3.1.2 Anchor shank, shackle, and shackle and crown pin. The anchor shank, shackle, and shackle and crown pin shall be forged steel in accordance with ASTM A668/A668M, class D, normalized, and tempered.

3.1.2.1 Test blocks. Test blocks for forgings shall be made from prolongs in accordance with ASTM A370. Specimens shall be taken at ¼T of prolong with the prolong having minimum dimensions of T by T by W, where T and W are thickness and width, respectively, of the largest cross section.

3.1.3 Stabilizers. Steel plates for stabilizers and gussets shall be in accordance with ASTM A36/A36M. Pipe shall conform to ASTM A53/A53M, type E, grade B, black, extra strong, in accordance with the following schedule:

Nominal pipe size (inches)	Schedule number
8	80
10	60
12	60
14	100
16	100
18	120

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3.2 Description.

3.2.1 Anchor. The anchor shall consist of a long shank and two sharp, double-swinging flukes. Anchors shall be furnished complete, including shackle and shackle pin with all parts properly fitted and assembled.

3.2.2 Stabilizer. The stabilizer shall consist of a half of steel pipe welded to the underside of a steel plate. This assembly, in turn, shall be welded to the underside of the anchor crown, on each side, with a gusset plate added for strength.

3.3 Design and dimensions. The anchor assembly, including shackle and shackle pin, shall comply with the design and dimensions shown on Drawing 803-860337 for type I anchors and Drawing 803-920808 for type II anchors. Stabilizers for type I, class 2 anchors shall be in accordance with the design and dimensions of Drawings 620656 or 3141650.

3.3.1 Mass tolerance. The mass tolerances for anchors shall not vary more than +4 percent to -2 percent of the nominal mass specified (see 3.3). The mass of the stabilizers, when used, shall not be included as part of the anchor nominal mass.

3.4 Magnetic particle.

3.4.1 Castings. For castings, all linear indications in accordance with MIL-STD-2035 that are over 6.35 mm (0.25 inch) in length shall be ground to sound metal and well faired with the surrounding surface. Weld repair shall not be required if remaining thickness of the original metal equals or exceeds the dimensions shown on the drawing (see 3.3). Depth of grinding shall be not greater than 38.1 mm (1.5 inches) except for cracks, which shall be removed fully. Castings may be built up to required dimensions by weld repair.

3.4.2 Forgings. Forgings shall comply with class 3 of MIL-STD-2035. Indications in excess of class 3 shall be ground to sound metal and well faired with surrounding surfaces.

3.4.3 Qualification of welders and welding procedure. Qualification of welders and welding procedure shall be accomplished in accordance with S9074-AQ-GIB-010/248. All final welds shall meet the requirements of MIL-STD-2035, class 2.

3.4.4 Repair welding. Unless otherwise specified (see 6.2), defects in castings and forgings may be removed and repaired by welding. Weld repairs shall use MIL-7018 electrodes in accordance with MIL-DTL-22200/l. Prior to repair welding, the excavated area shall be fully visible to the welder and allow access of the electrode to all weld surfaces.

3.4.4.1 Minor weld repairs. Minor weld repairs may be made to correct surface defects provided that:

a. they do not exceed 25 mm (0.98 inch) in depth or 20 percent of material thickness, whichever is less

b. a weld buildup for correction of dimensional or machining errors does not exceed the following limitations:

(1) 4.76 mm (0.187 inch) maximum buildup for design thickness 25.4 mm (1 inch) and under

(2) 20 percent maximum buildup for design thickness over 25.4 mm (1 inch), but not to exceed 9.52 mm (0.375 inch) unless approved

(3) contouring (such as grinding or machining) for discontinuities shall blend into adjacent material and shall not reduce the thickness of the material below minimum design requirements

3.4.4.2 Nominal weld repairs. Repair welds in excess of those specified as minor repairs shall be made only with the approval of the Government representative.

3.5 Proof load. Anchors shall comply with proof load test specified in 4.5.3. There shall be no permanent deformation of any part of the anchor upon completion of this test.

3.6 Cleaning. The castings and forgings shall be cleaned for proper adhesion of finish when offered for inspection.

3.7 Identification markings. The crown shall be marked in conformance with the applicable drawing (see 3.3).

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3.8 Finish. Surfaces shall be cleaned in accordance with SSPC-SP 1 to remove oil, grease, and residual contaminants. Anchor surfaces shall be prepared with near white metal blast in accordance with NACE No. 2/SSPC-SP 10. Waterjetting to NACE WJ-2/SSPC-SP WJ-2 may be used in place of NACE No. 2/SSPC-SP 10. Anchors shall have a primer of one coat in accordance with MIL-PRF-23236, type V or VI, class 5 or 7, 4 to 8 mils; or one coat in accordance with MIL-PRF-23236, type VII, class 5 or 7, 4 to 8 mils. Anchors shall have a top coat of one coat in accordance with MIL-PRF-23236, type V or VI, class 5 or 7, 4 to 8 mils; or one coat in accordance with MIL-PRF-23236, type VII, class 5 or 7, 10 to12 mils. Anchors shall have one coat of haze gray in accordance with MIL-PRF-24635; type III, V, or VI; class 1; grade B or C; 2 to 3 mils.

3.9 Stabilizer welding. Stabilizers shall not be welded to the anchor assembly until both have arrived at their destination.

3.10 Workmanship. The workmanship shall be free from imperfections that may impair appearance or serviceability.

4. VERIFICATION

4.1 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the tests of 4.5.

4.2 Sample lots.

4.2.1 Castings. A lot shall consist of castings made from the same heat. Where material cannot be identified by heat, it shall be divided into lots of not more than 500 kg (1,000 lbs) each, except where a single casting has a mass of more than 500 kg.

4.2.2 Forgings. A lot shall consist of material manufactured from the same heat.

4.2.3 Chemical analysis of forgings. A lot shall consist of all forgings from the same heat.

4.2.4 Chemical analysis of castings. A lot shall consist of castings of one class made from the same heat, heat-treated in the same furnace charge, and offered for inspection at one time.

4.2.5 Assembled anchor. A lot shall consist of assembled anchors of the same size offered for inspection at the same time.

4.3 Sampling for lot acceptance.

4.3.1 Sampling for visual and dimensional examination. Sampling for visual and dimensional examination and lot acceptance criteria shall be in accordance with table II.

TABLE II. Sampling for visual and dimensional examination.

Number of anchors in inspection lot	Number of anchors in sample	Acceptance number (defectives)
25 and under	10	0
26 to 40	15	1
41 to 110	25	1
111 to 300	35	2
301 to 500	50	3
501 to 800	75	4
801 and over	110	6

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4.3.2 Sampling for nominal mass. Sampling for nominal mass examination and lot acceptance criteria shall be in accordance with table III.

TABLE III. Sampling for nominal mass tests, assembled anchors tolerance.

Number of anchors in inspection lot	Number of anchors in sample	Acceptance number (defectives)
40 and under	5	0
41 to 110	10	0
111 to 300	15	1
301 to 500	25	1
501 to 800	35	2
801 and over	50	3

4.3.3 Sampling for magnetic particle test. Sampling for magnetic particle tests shall be 100 percent for all parts except that pins shall not be subjected to this test.

4.3.4 Sampling for chemical analysis of castings. Where material cannot be identified by heat, it shall be divided into lots of not more than 500 kg (1,000 lbs). One sample shall be obtained from each lot for chemical analysis and these samples shall be analyzed individually.

4.3.5 Sampling for chemical analysis of forgings. Samples for chemical analysis of forgings shall be taken from the same lot that consists of material manufactured from the same heat.

4.3.6 Sampling for tension test and microstructure tests on castings. For castings estimated to have a mass of 250 kg (500 lbs) and over, one sample for the tension test shall be taken to represent each casting. For castings estimated to have a mass less than 250 kg (500 lbs), two samples for the tension test shall be taken from each lot. Microstructure examination for castings shall be performed on a specimen taken from the test block adjacent to the tensile specimen and away from the surface of the test block.

4.3.7 Sampling for tension tests on forgings. For all forged parts, except shackles and pins from anchors having a mass of 500 kg (1,000 lbs) and less, two samples for the tension test shall be taken from each lot.

4.4 Examinations.

4.4.1 Visual and dimensional examination. Anchors selected in accordance with table II shall be visually and dimensionally examined in accordance with Drawing 620656 or 3141650, and when applicable, their stabilizers shall be examined for defects in manufacturing and workmanship to verify compliance with this specification both prior to and after painting. Stabilizers shall not be subjected to any examinations, tests, analyses, or inspections except those related to welding and welding procedure (see 3.4.3) and visual and dimensional examination. Any anchor containing one or more visual or dimensional defects shall be rejected and if the number of defective anchors in any sample exceeds the acceptable number for that sample, the entire lot represented by the sample shall be rejected. Rejected lots may be offered again for inspection, provided that all nonconforming anchors have been removed. Samples shall again be selected from such rejected lots and re-examined to verify compliance with this specification.

4.4.2 Shank. Radii on the corners of the shanks shall be examined to ensure a smooth curve, free from irregularities. The stop on the shank shall make full surface contact with the boss in the crown when the flukes are in the operating position.

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4.4.3 Determination of mass. Each anchor of the sample shall have its mass determined to verify compliance with 3.3. Anchors exceeding the tolerance limit specified on the drawing (see 3.3) shall be rejected. If the number of defective anchors in any sample (see table III) exceeds the acceptable number for that sample, the entire lot represented by that sample shall be rejected. Rejected lots may be resubmitted for inspection only after all nonconforming anchors have been removed. Samples shall again be selected from such rejected lots and re-examined to verify compliance with this requirement.

4.5 Test procedures.

4.5.1 Chemical tests. Samples selected in accordance with 4.3.4 or 4.3.5 shall be analyzed in accordance with ASTM A27/A27M or ASTM A668/A668M to ensure conformance with chemical composition requirements. The methods shall be correlated with the National Institute of Standards and Technology standard reference material to ensure the validity of the method that is used as a control in chemical methods of analysis or for calibration in instrumental methods of analysis. The specification covering the method of analysis used shall be as specified (see 6.2).

4.5.2 Mechanical property tests. Tension tests shall be conducted in accordance with ASTM A370.

4.5.3 Proof load test.

4.5.3.1 General. Each anchor shall be assembled with shackle and pins, ready for service, and shall be subjected, in the presence of the Government representative, to the proof load test in both positions as specified on the applicable drawing (see 3.3).

4.5.3.2 Gaging. The anchors shall be gaged before and after they have been tested in each operative position to determine whether any permanent deformation has been caused by applied stress.

4.5.4 Microstructure test. Microstructure examination shall be in accordance with ASTM E112.

4.5.5 Magnetic particle test. Magnetic particle examination shall be performed following completion of proof load testing and shall be in accordance with T9074-AS-GIB-010/271. The anchor proper and all components of the anchor assembly, except stabilizers, shall be subjected to magnetic particle examination. Examinations shall be performed after proof loading on finished components and prior to any painting. Special attention shall be given to the following:

a. fluke surfaces from tip up to and including radii at junction with crown
b. crown pin lug surface including fillets at junction of lug with crown and interior of hole for shank
c. crown bearing area that acts as a stop for the shank
d. shackle pin area and “lower” bearing surfaces, both on the shank
e. the shackle at bend and pin hole
f. all welds in the anchor assembly

Repair welds shall be subjected to the same inspection standards required of the casting and the area shall be suitably marked to facilitate inspection. Anchors which are repair welded shall be proof load tested afterwards.

Any part in the sample containing one or more defects shall be rejected. Rejected lots may be resubmitted for inspection only after every anchor part has been examined for the type of defect found and all defective parts have been removed from the lot.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activities within the Military Service or Defense Agency, or within the military service’s system commands. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Anchors without stabilizers are for use on ships. Stabilizers are used on anchors in connection with mooring systems. Navy stockless anchors made to this specification are a design with known performance characteristics and geometric properties that are compatible with U.S. Navy ships. Existing stock of anchors manufactured to DOD-A-24581 dated 25 October 1978 is acceptable for use until depleted.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and class of anchor desired (see 1.2.1).

c. Nominal mass of anchor required (see 1.2.2).

d. Repair welding if other than specified (see 3.4.4).

e. The specification covering the method of analysis used in chemical tests (see 4.5.1).

f. Packaging requirements (see 5.1).

6.3 Applicability. Anchors covered by this specification are commonly referred to as Navy stockless anchors. Type II anchors are also known as MK II Navy stockless.

6.4 Subcontracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for use or incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Subject term (key word) listing.

Crown

Crown pin

Fluke

Shackle

Shank

Stabilizer

Test block

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Preparing activity:
Navy – SH
(Project 2040-2019-001)

Review activity:
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